SOLUTIONS

Module:	Embedded Systems		
Module Code	EBU5477	Paper	A
Time allowed	2 hours	Filename	**Solutions_2324_EBU5477_A**
Rubric	ANSWER ALL FOUR QUESTIONS		
Examiners	Dr Matthew Tang	Dr Jin Zhang	

Solutions

Question 1

a) Imagine that you are an embedded system designer in an electronic consumer product company. You are given a context diagram of the design of a Bluetooth speaker, as shown in **Figure 1**. Study the diagram carefully and answer the following questions.

[15 marks]

i) Based on the information provided in **Figure 1**, sketch a hardware block diagram for the speaker. Your diagram should focus on the physical objects in the embedded system. For each connection of the blocks, identify a possible hardware interface (e.g. GPIO, ADC, etc.).

(5 marks)

ii) Sketch an organigram that illustrates the hierarchy of the embedded software for the speaker. You should make sure that the information is consistent with your answer in **(i)**.

(5 marks)

iii) One of your colleagues suggests that using recycled aluminium for the case of the speaker can improve the durability of the product by 50% with an increase of the cost of the whole product for 25%. **In your opinion**, what would be the impact to the environment if this suggestion is accepted? Hence, state whether you agree or disagree to the suggestion.

(5 marks)

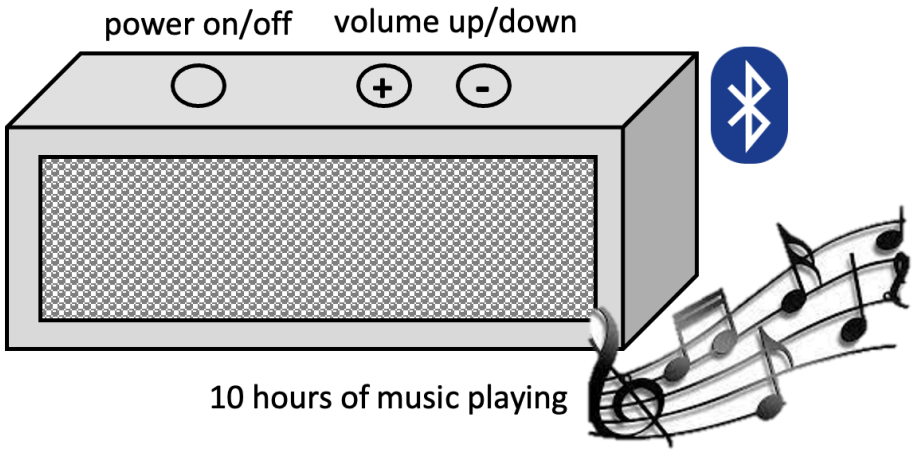


Figure 1: Context Diagram of a Bluetooth Speaker

SOLUTION

(i) (This question is open-ended so different kinds of sketch are acceptable)
Marks are allocated to the following features in the diagram.
(1 mark) a processor block
(1 mark) a button for power on with connection GPIO
(1 mark) a button(s) for volume up and down with proper connection GPIO
(1 mark) a Bluetooth module with connection I^2C/SPI/GPIO

(1 mark) a speaker with connection DAC/I^2A
Other reasonable blocks like battery can substitute all but the processor block.

(ii) (This question is open-ended so different kinds of sketch are acceptable)
Marks are allocated to the following features in the diagram.
(1 mark) main at the top
(2 marks) ANY TWO of music playing, volume control, Bluetooth/network connection (first level)
(2 marks) ANY TWO physical blocks from (i) connected correctly, e.g. Bluetooth module with Bluetooth connection; speaker with music playing

(iii) **(2 marks)** The use of recycled aluminium first reduces the usage of world's natural resources.
(2 marks) The longevity of product means that people are less likely to replace them and fewer waste will be created.
(1 mark) As both points above are positive in a perspective for the environment, I would **agree** with the suggestion.
(*it is acceptable to disagree but the reasons should support the decision.*)

b) Two ARM Cortex-M4 instructions are shown below:

```
MOV r0, r1
ADD r2, r0, r1
```

It is known that r0 = 4, r1 = 5, r2 = 6 before the instructions were executed.

[5 marks]

i) What are the final values in the registers r0, r1 and r2 respectively after the execution?

(3 marks)

ii) What would be the total size (in bytes) of the two instructions if both are very common?

(2 marks)

SOLUTION

(i) **(1 mark)** r0 = 5
(1 mark) r1 = 5
(1 mark) r2 = 10
(ii) **(2 marks)** Total size = 2 x 2 = 4 bytes

c) An ARM Cortex-M4 microcontroller maps all its physical SRAM to the memory addresses between 0x20000000 and 0x2001FFFF. Calculate the size (in bytes) of the physical SRAM.

SOLUTION

The size of the physical SRAM
= 0x2001FFFF – 0x20000000 + **1 (2 marks)** bytes **(1 mark)** (*each address refers to a byte*)
= 0x20000 bytes
= 2^{17} bytes **(1 mark)**
= 128 kB **(1 mark)**

Solutions

Question 2

a) The snippet of a main program for an ARM Cortex-M4 processor is shown in **Figure 2**.

[10 marks]

```
int func1(int a, int b){
    int x, y;
    x = a * a + b * b;
    y = sqrt(x); // square root function from math.h
    return y;
}
void main(void){
    …
    d = func1(3, 6);
    …
}
```

Figure 2: Code snippet of an Cortex-M4 main program

i) According to AAPCS, what registers should be used to pass arguments *a* and *b* from **main()** to **func1()?**

(2 marks)

ii) Integers *x* and *y* inside **func1()** are called automatic variables. Explain how memory space would be allocated for these two variables.

(3 marks)

iii) What must be done in the beginning of **func1()** such that the program execution could return to **main()** eventually? And why?

(3 marks)

iv) How is the answer *y* from **func1()** returned back to **main()**?

(2 marks)

SOLUTION

(i) **(2 marks)** *a* and *b* are passed in registers r0 and r1 respectively.

(ii) **(2 marks)** 4 bytes on the stack (area of SRAM) will be allocated for each of the variables **(1 mark)** by subtracting the value in register SP by 8 (i.e. 8 bytes).

(iii) **(2 marks)** The return address in register LR has to be pushed to the stack and popped back to LR before returning to main().
(1 mark) This is because the value will be destroyed when the call is made to sqrt().

(iv) **(2 marks)** The return value is saved in register r0.

Solutions

b) A snippet of a C program for an ARM Cortex-M4 processor is shown in **Figure 3**. It works with the LCD display and the UART hardware using the standard drivers as in our lab. exercises.

Explain briefly the meaning of the statements at

(i) line 2; (ii) line 4, (iii) line 7, and (iv) lines 08-11.

[5 marks]

```
01: void main_test(void){
02:     uart_init(9600);
03:     uart_enable();
04:     lcd_init();
05:
06:     while (1) {
07:         char c = uart_rx();
08:         if (c == '0')
09:             lcd_clear();
10:         else
11:             lcd_put_char(c);
12:     }
13:}
```

Figure 3: Code snippets of a Cortex-M4 program for LCD and UART

SOLUTION

(i) **(1 mark)** Initialise the UART hardware to a BAUD rate of 9600 bit/second

(ii) **(1 mark)** Initialise the LCD display (and get ready to print)

(iii) **(1 mark)** Wait for and receive a character from UART

(iv) **(1 mark)** if the received character is '0', clear the LCD display, otherwise
(1 mark) print the received character to the LCD display.

c) Consider a system built on a 48 MHz Cortex-M4 processor with a main loop and only a single interrupt which occurs at a 30 kHz frequency. Assume that the ISR takes 600 clock cycles to execute and there is a total of 12 clock cycles of response and return time overhead.

[10 marks]

i) What percentage of the processor's time (utilisation) is spent in the ISR and the overhead? So what is the percentage of the processor's time left for the main loop to execute?

(4 marks)

ii) If the main loop requires 55 ms of computation to execute one iteration, what is the maximum main loop update rate (in Hz)?

(2 marks)

iii) State two benefits of interrupt over polling

(4 marks)

Solutions

SOLUTION

(i) Percentage of the processor's time in ISR
= 30 x 10^3 x (600 + 12) x (1 / 48 x 10^6)x 100% **(1 mark)**
= 38.25% **(1 mark)**
Percentage of the processor's time for the main loop
= 1 – 38.25% **(1 mark)**
= 61.75% **(1 mark)**

(ii) Max. main loop update rate
= 61.75% x 1 / (55 x 10^{-3}) **(1 mark)**
= 11.23 Hz **(1 mark)**

(iii) **ANY TWO** of the following **(2 marks each, 4 marks in total)**:

- Interrupts are generally more efficient as the CPU can focus on other tasks while waiting for external events.
- Low latency. Events are serviced immediately after they raise an interrupt request
- Fast response: interrupts enable immediate responses to external events where timely reactions are critical
- Prioritisation: Interrupts can be prioritised programmatically, allowing the system to respond to the most important events first.

Question 3

a) a) Two GPIO ports are used to demonstrate the addition of two bits. GPIO_A is used to configure two GPIO pins (Pin 1: SW1 and Pin 2: SW2) to read the status of two different switches. GPIO_B is used to configure two GPIO pins (Pin 1: LED1 and Pin 2: LED2) are used to control two LEDS based on the status of the switches, **as in Table 1**.

[15 marks]

Table 1: SW and LED relationship

	SW1	SW2	LED1	LED2
Initial state	**0**	**0**	**0**	**0**
	0	1	0	1
	1	0	0	1
	1	1	1	0

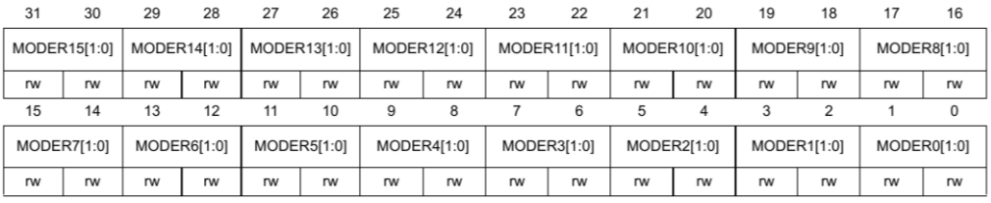


Figure 4: MODER register

Solutions

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Figure 5: IDR and ODR registers

i) Write one possible value of MODER_A and MODER_B (see **Figure 4**).

- 00: Input (reset state)
- 01: General purpose output mode
- 10: Alternate function mode
- 11: Analog mode

Hint: It is enough to show the first word of MODER (bit 15 to bit 0), e.g., for MODER with 0 everywhere, you can write 0000

(4 marks)

ii) Write one possible value of IDR_A, IDR_B, ODR_A, and ODR_B when SW1=1 after the initial state (see **Figure 5**)

(6 marks)

iii) Write one possible value of IDR_A, IDR_B, ODR_A, and ODR_B when SW2=1 after ii) above.

(5 marks)

SOLUTION:

i) Student can show one word (bit15-Bit0), X could be anything, {…} is the list of correct values
MODER_A: X X {0,4,8,C} {0,1,2,3} (2 marks)
MODER_B: X X {1,5,9,D} {4,5,6,7} (2 marks)

ii) IDR_A: X X X {2,3,6,7,A,B,E,F} (2 marks)
IDR_B:X X X X (1 mark)
ODR_B:X X X X (1 mark)
ODR_B: X X X {2,3,6,7,A,B,E,F} (2 marks)

iii) IDR_A: X X X {6,7, E, F} (2 marks)
IDR_B:X X X X (0.5 mark)
ODR_B:X X X X (0.5 mark)
ODR_B: X X X {4,5,6,7,C,D,E,F} (2 marks)

Solutions

b) This question relates to asynchronous serial communication as shown in **Figure 6**.

[4 marks]

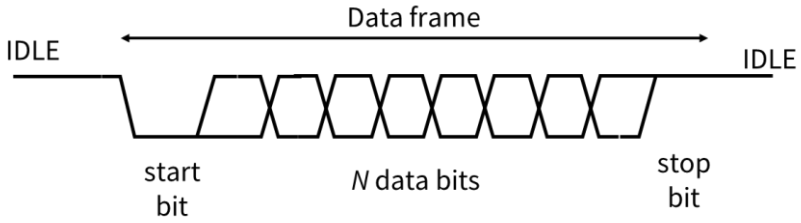


Figure 6: UART Data Frame

The following data frame is transmitted: 1010101X, where the last bit (X) is a parity bit. Answer the following questions:

i) What is the value of N in this case?

(1 mark)

ii) What is the value of X based on the received decisions in **Table 2**. Is it Odd or Even parity?

(3 marks)

Table 2: Parity Decision Table

Received data frame	**Decision**
10101110	No Error
00101011	Error

SOLUTIONS:

i) N = 8 bits (1 mark)

ii) X = 1 (2 marks) and Odd parity (1 mark)

c) This question relates to the USART baud rate setting for an $f_{PCLF} = 12MHz$ and a desired Baud rate of 460.8Kbps. Note that *OVER8*=0 indicates oversampling by 16 and *OVER8*=1 indicates oversampling by 8. The relation between f_{PCLF} and the actual baud rate is shown in **Figure 7** and depends on the setting of *OVER8* and *USARTDIV*. The value of *USARTDIV* can be obtained based on *DIV_Mantissa* and *DIV_Fraction* in **Figure 7**.

[6 marks]

$$T_x\backslash R_x\ (baud) = \frac{f_{PCLF}}{8 \times (2 - OVER8) \times USARTDIV}$$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV_Mantissa[11:0]												DIV_Fraction[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Figure 7: Baud rate equation and the DIV register

i) What should be the value of *USARTDIV* and *DIV_Fraction* for oversampling by 16?

(4 marks)

Solutions

ii) What would be the error between the desired Baud rate and the actual one?

(2 marks)

SOLUTIONS:

i) USARTDIV=12000/(16x460.8)=1.6276 (1mark);
DIV_Mantissa= 0 0 1 (1 mark)
DIV_Fraction= 10/16= 0.625 (1 mark)
USARDIV=1.625 (1 mark)

ii) Actual baud rate: 12MHz/(16 x 1.625) = 461.532 kbps (1 mark)
Error = (461.532-460.8)/ 461.532=0.1586=> 0.16% (2 mark)

Question 4

a) Suppose you are going to flash an LED on and off periodically as shown in **Figure 8**, using delay by software or hardware timer. The processor clock frequency is 84 MHz. Answer the following questions.

[12 marks]

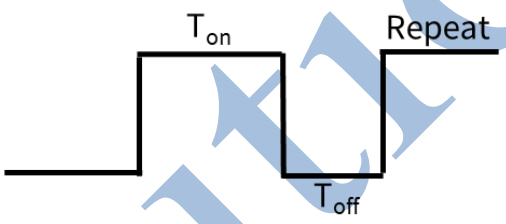


Figure 8: Periodic pulse for LED flashing

i) A student wrote the following loop for a time delay. Do you think this is a reasonable method? Explain briefly

```
for (i = 0; i < 1000; i++);
```

(4 marks)

ii) Using the SysTick timer to create the pulses, calculate the initial value for the SysTick timer if you want to count for 50 µs.

(4 marks)

iii) If T_{on}=150 µs, T_{off} =100 µs, based on (ii), suggest how to use SysTick timer to get a delay for the on/off states.

(4 marks)

SOLUTION:

i) YES, when there is no interrupt, the loop executes in a near constant time.
OR
NO, the runtime for the loop is affected by the asynchronous interrupt.
(2 marks) YES or NO (only with good justification) (2 marks) reasonable justification.

ii) $Clock\ cycle = \frac{1}{84M} = 11.9\ \text{ns}$

Solutions

$$Initial\ value = \frac{50 \times 10^{-6}}{11.9 \times 10^{-9}} = 4201\ (or\ 4202)$$

iii) Set the initial value of SysTick timer using the value in ii). (2 marks)
Then with a variable, count the high time for 3 x 50 µs and low time for 2 x 50 µs. (2 marks)

b) A Cortex M4 microcontroller (MCU) connects to 3 LCD displays using the I^2C bus. The I^2C slave addresses for the displays are 0x28, 0x29, and 0x2A respectively. Answer the following questions.

[13 marks]

i) Can you connect to more LCD displays, and why?

(3 marks)

ii) Draw a diagram to show the electrical signals (with high and low levels) on the I^2C buses when the MCU tries to address and write to the LCD of address 0x28 successfully.

(7 marks)

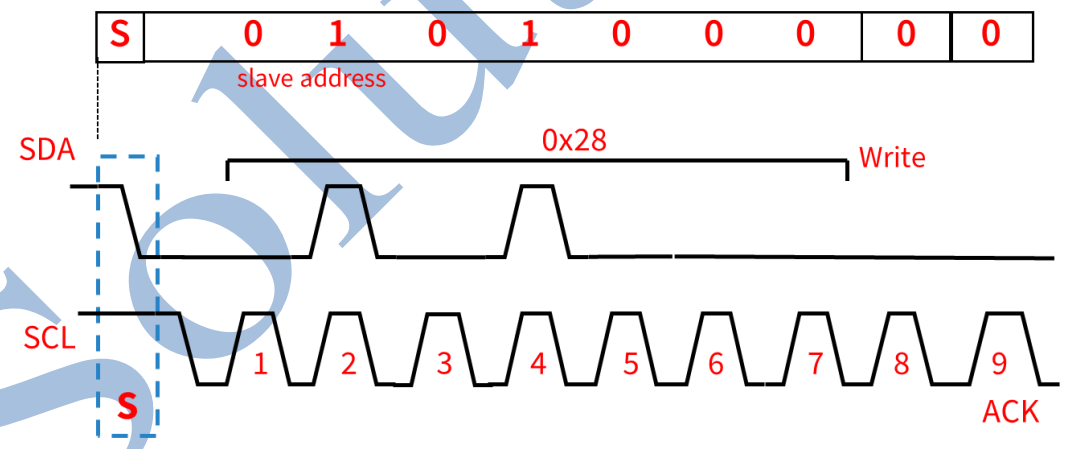
iii) When two (or more) masters (M1 & M2) begin a transfer at the same time, what technique can be used to decide which master can get the I^2C bus access?

(3 marks)

SOLUTION:

i) Yes (1 mark), because you can connect up to 127 slave devices on the same bus (2 marks)

ii)



Digital bits table (2 marks)
Start bit (S) (1 mark)
Sensor address 0x2A (2 marks) (consider also SCL)
Read/Write bit (1 mark)
Acknowledge bit (from LCD) (1 mark)

iii) Bus arbitration (3 marks)

- End of Solution -

Solutions